



Iceberg Auto Table Maint ***Starburst Office Hours***

An interactive webinar
with Lester Martin from DevRel

Connection before content

Lester Martin – <https://linktr.ee/lestermartin>

- Developer Relations @ Starburst
 - Blogging & forums
 - Webinars & videos
 - User groups & events
 - Training & tutorials
- 30+ years of technology experience
 - Started journey on TRS-80 Model III
 - Played most roles, but a programmer at my core
 - ½ career in OLTP and ½ in data analytics
 - Decade+ of “big data” experience to include
 - Trino/Starburst, Hadoop, Hive, Spark
 - NiFi, Kafka, Storm, Flink
 - HBase, MongoDB

devrel@starburst.io

Starburst is an Open, Hybrid Lakehouse platform



Analytics Accelerators

Data Products, Warp Speed, and other analytics productivity tools.

Query Engine

powered by 🐰 trino

Enterprise Platform

Enterprise-grade security, scalability, governance, usability.

Data Connectors & Ingestion

Fast, easy data access, including Iceberg table creation

 Starburst Enterprise

 Starburst Galaxy

 AIDA

+ New chat

Persona

Analyst

Cluster

aws-us-east-2

Data product

Air & Space

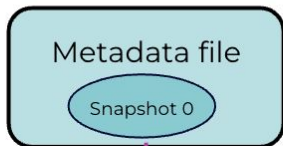
[View data product](#)

Architecture overview

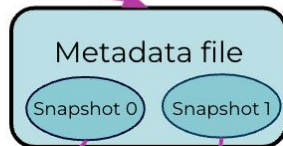


snapshots are created anytime data or structure is changed

Metadata layer
lives in the
./metadata dir



Data files



Data files

saved in the
./data dir

Data layer



Data files

metadata and data persisted together on the data lake

Iceberg Metadata Tables & Columns

\$properties - Show basic information about the table

\$history - Shows basic history of the table

\$snapshots - Shows history of snapshots

\$manifests - List of manifest files

\$partitions - Information on partitioning on the table

\$files - Data files associated with the table

\$path - Full file system path name of the data file for this row

\$file_modified_time - Timestamp of the last modification of the data file for this row

```
SELECT * FROM "<table_name>$properties"
```

```
SELECT *, "$path", "$file_modified_time" FROM <table_name>
```

Iceberg Metadata Tables - Example

committed_at	snapshot_id	parent_id	operation	manifest_list	summary
2022-06-02 07:00:39.747 ...	8898509898101371000	NULL	append	s3://lakehouse-sb/blog/iceberg_test_snaps...	{ changed-partition-count ...
2022-06-02 07:00:42.923...	6576729567558493000	8898509898101371000	append	s3://lakehouse-sb/blog/iceberg_test_snaps...	{ changed-partition-count ...
2022-06-02 07:00:46.201 ...	8373865355217030000	6576729567558493000	append	s3://lakehouse-sb/blog/iceberg_test_snaps...	{ changed-partition-count ...
2022-06-02 07:00:49.283...	1978685887502723000	8373865355217030000	append	s3://lakehouse-sb/blog/iceberg_test_snaps...	{ changed-partition-count ...
2022-06-02 07:00:52.276 ...	3470249778653180400	1978685887502723000	append	s3://lakehouse-sb/blog/iceberg_test_snaps...	{ changed-partition-count ...

```
SELECT * FROM mycatalog.myschema."<table_name>$snapshots"
```

Iceberg Metadata Deep-Dive

<https://www.starburst.io/community/forum/t/iceberg-metadata-blog-series/>

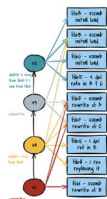
W Lester Martin (l11n) – 27 Dec 24



iceberg acid transactions with partitions (a behind the scenes perspective) ²

a port of my prior post taking a deeper look at what happens under the hood of hive with “acid” transactions — this time on iceberg tables with parquet files

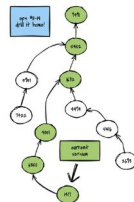
W Lester Martin (l11n) – 7 Jul 24



iceberg snapshots affect storage footprint (not performance) ¹

it is easy to understand why most folks initially imagine that iceberg’s ability to maintain a long history of snapshots will cause performance problems, but that is not the case — the ...

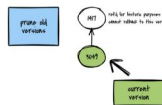
W Lester Martin (l11n) – 16 Feb 24



iceberg snapshot is_current_ancestor flag (what does it tell us)

i’ve noticed the is_current_ancestor column of the apache iceberg \$history metadata table for a while now – it wasn’t until I got a direct question about it that i realized it was...

W Lester Martin (l11n) – 21 Feb 24



apache iceberg table maintenance (is_current_ancestor part deux) ²

as a follow-on to my earlier post about iceberg versioning (and the is_current_ancestor flag), i thought it would be useful to show working examples of the maintenance activities that are needed to...

Video metadata walkthrough at https://www.youtube.com/watch?v=jnb_2Nlct34

More of my Iceberg ramblings at <https://lestermartin.blog/tag/iceberg/>

Data maintenance jobs

Automated solutions with configurable options

 Starburst Galaxy

< >

Query

^

Query editor

Query insights

Saved queries



Data

^

Catalogs

Data ingest

Data maintenance

Data products

Data quality



AI models

Configure data maintenance

Schedule maintenance tasks for  Iceberg tables [🔗](#)



All schemas

Includes newly created schemas. Dropped schemas and those with individual schedules are excluded.

Maintenance tasks



Compaction

Optimize data file sizes to improve performance.



Profile & statistics

Collect statistics about your data to improve query performance.



Snapshot expiration

Free up storage by deleting outdated data snapshots.

Threshold *

7

days



Delete orphan files

Free up storage by deleting unused data files.

Select cluster *

Job schedule

Time zone *



Select frequency



Enter cron expression

Cron expression *

0 ****



Demo